



A real-time image dehazing algorithm based on adaptive routing

Kun Meng¹ · Shengjie Yang¹ · Zhen Liu¹ · Sheng Li² · Min Wang¹ · Haibin Liao¹ · Li Yuan¹

Accepted: 2 June 2025

© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2025

Abstract

Most image dehazing algorithms in practical applications suffer from weak input adaptability due to their neglect of input image types (hazy or haze-free), and they often enhance dehazing performance by deepening network layers, resulting in large parameter counts that make real-time processing difficult. To address these issues, we propose a real-time image dehazing algorithm based on adaptive routing. Firstly, we design an adaptive routing module based on multi-scale feature extraction to determine the processing path of the input image. Then, based on the traditional feature attention (FA) module, a lightweight de-redundant feature attention (DRFA) module is constructed for dehazing processing. Finally, the residual network with the PixelShuffle upsampling module is utilized for efficient image recovery. The experimental results demonstrate that our algorithm achieves high PSNR and SSIM scores on several datasets, and we reach an average PSNR and SSIM of 26.75 dB and 0.832 on the mixed dataset of clear and hazy images, respectively. These results outperform other comparative algorithms, indicating the robustness and practicality of our approach. Not only that, our algorithm has a computational cost of 8.93G FLOPs and a dehazing time of only 4.81 ms during the 256×256-pixel image dehazing efficiency test, which meets the requirements of real-time processing for video surveillance in outdoor scenes.

Keywords Image dehazing · Deep learning · Adaptive router · PixelShuffle · Lightweight

✉ Li Yuan
2012015@wtu.edu.cn

¹ School of Electronic and Electrical Engineering, Wuhan Textile University, Wuhan 430200, China

² The University of Edinburgh, Edinburgh EH8 9YL, UK

1 Introduction

In hazy weather, light scattering caused by tiny particles degrades image quality, affecting clarity and visual perception, which can interfere with the performance of subsequent computer vision tasks, making image dehazing techniques crucial [1–3]. The traditional image dehazing algorithms mainly include image enhancement-based dehazing algorithms and prior-based dehazing algorithms. Among them, image enhancement-based dehazing algorithms can lead to color distortion and missing information in dehazed images because they do not consider the imaging mechanism of hazy weather. The prior-based dehazing algorithms usually use the atmospheric scattering model (ASM) [4] as their theoretical basis and combine it with handcraft priors to remove haze from the image. Such algorithms have achieved remarkable results in image dehazing, but they are overly dependent on specific scenes and prior assumptions, which leads to the limitation of their generalization ability and robustness. In recent years, with the rapid development of deep learning, deep learning-based dehazing algorithms have become the mainstream in this field. These algorithms use convolutional neural networks (CNN) to construct the dehazing network and achieve ideal dehazing results. Despite the significant advantages of such dehazing algorithms, they still face the problems of weak model input adaptation and high computational cost in outdoor scenes. To address the problems above, we propose a real-time image dehazing algorithm based on adaptive routing, with the following contributions:

- Aiming at the problems of poor robustness and adaptability of current dehazing algorithms in outdoor scenes, an adaptive routing module based on multi-scale feature extraction is constructed. The module analyzes the multi-scale downsampled feature maps and scores the features of the input images in combination with the adaptive routing mechanism to decide whether the input images need to be dehazed, to improve the robustness and adaptability of the dehazing algorithm in outdoor scenes while saving unnecessary computational costs.
- We design a lightweight de-redundant feature attention (DRFA) dehazing module to improve the dehazing algorithm's efficiency by eliminating redundant information in low-resolution features. The module is based on the traditional feature attention (FA) module and introduces spatial and channel squeeze convolution (SCConv) to reduce feature spatial and channel redundancy. At the same time, the module uses residual concatenation to preserve the key feature information during the dehazing process.
- To reduce the artifacts and checkerboard effects that occur during the upsampling process and improve the dehazing visual quality of our algorithm. We propose an upsampling module that combines a residual network and PixelShuffle. This module uses PixelShuffle to increase the image resolution and employs a convolutional layer to learn mapping and reconstruction information, achieving a high-quality output of reconstructed images with a relatively low number of parameters and computational cost.

The rest of the paper is as follows: The second section briefly reviews the methods related to image dehazing, the third section describes the proposed network composition in detail, the fourth section shows the experimental results and evaluation, and the fifth section discusses and draws the conclusion.

2 Related work

2.1 Traditional image dehazing algorithms

Early image dehazing techniques mainly rely on image enhancement algorithms to improve the contrast and clarity of images, thereby enhancing the visual quality of images. Commonly used algorithms include those based on histogram equalization, Retinex theory, and homomorphic filtering. For example, Xu et al. [5] proposed a dehazing algorithm based on contrast-limited adaptive histogram equalization, which can suppress noise while enhancing image contrast and improving the dehazing effect. Jobson et al. [6] proposed the multi-scale Retinex with color restoration (MSRCR) algorithm. By defining a color restoration method to correct image defects, it can restore high-definition images in low-light environment. Yu et al. [7] proposed a dehazing algorithm based on homomorphic filtering, which can not only improve the degradation of images but also effectively enlarge image details and enhance image contrast. The above dehazing algorithms have the advantages of simplicity and efficiency. However, ignoring the degradation factors of images will lead to problems such as loss of local image information and unsatisfactory dehazing effects.

To more accurately restore clear images, researchers base their methods on the atmospheric scattering model, using reasonable assumptions and prior knowledge to estimate and reconstruct dehazed images. For example, He et al. [8] proposed a dark channel prior (DCP) algorithm, which estimates the atmospheric light and scene transmission in an image to remove haze, achieving effective dehazing results to some extent. Zhu et al. [9] introduced a color attenuation prior (CAP)-based dehazing algorithm that estimates the distribution of color attenuation caused by haze in images and then corrects the color attenuation effect to restore the color. Berman et al. [10] presented a non-local prior-based dehazing algorithm that utilizes statistical texture information from natural images to analyze and estimate the transmission of an image, allowing for better adaptation to different scenes. Although prior-based dehazing algorithms have achieved certain results, they rely heavily on specific scenes and prior assumptions, leading to poor generalization and robustness.

2.2 Deep learning-based dehazing algorithms

With the continuous progress of deep learning technology, deep learning-based algorithms have emerged in recent years to solve the problem of image dehazing. Cai et al. [11] proposed a DehazeNet network, which can estimate the medium transmittance of an image and bring it into the physical imaging model to obtain the restored

image. The AOD-Net dehazing algorithm proposed by Li et al. [12] deforms the atmospheric scattering model, combines atmospheric light and transmittance into one variable, and then generates a dehazed image through a convolutional neural network. The above algorithms are all based on the atmospheric scattering model to perform dehazing processing on images, but still cause problems such as color distortion due to the limitations of the physical model. The direct end-to-end dehazing algorithm no longer relies on the physical model and restores the clear image by learning the mapping between hazy images and clear images. For example, Dong et al. [13] proposed a multi-scale boosted dehazing network with dense feature fusion network (MSBDN). By developing a simple and effective enhanced decoder, the clear image is gradually restored. Hong et al. [14] proposed a knowledge distilling dehazing network (KDDN), which extracts image dehazing through heterogeneous task simulation. Guo et al. [15] proposed the DeHamer dehazing network, which enhances the dehazing performance by learning a modulation matrix based on Transformer features to modulate CNN features and introducing a prior related to haze density.

Although deep learning-based dehazing algorithms exhibit excellent performance in dehazing effects, they may still face issues such as input adaptability and find it difficult to meet the usage requirements of outdoor scenes. To further enhance the generalization and adaptability of dehazing network models, Wang et al. [16] proposed an end-to-end haze concentration adaptive network (HCAN). This network uses a pyramid feature extractor to obtain features of different layers and, through the feature enhancement module, adaptively perceives and integrates hazy images of different concentrations, ultimately restoring a clear image. Yu et al. [17] proposed an adaptive multi-information distillation dehazing network. Through a multi-information distillation module, it adaptively assigns weights to different haze regions and gradually extracts important features to restore different types of hazy images. Li et al. [18] proposed an attention-based adaptive feature selection for multi-stage image dehazing network (ASNet). By adaptively selecting and aggregating, it extracts rich detailed features to restore hazy images with different concentrations and uneven distributions. Although these algorithms possess certain adaptive capabilities, they still suffer from issues such as over-processing for clear input images, and their high parameter count and latency significantly restrict their application in real-time scenarios.

3 Proposed methods

3.1 Network architecture

The overall framework of the algorithm is shown in Fig. 1. First, the features of the input image are scored through the adaptive routing module to determine whether the image needs to enter the dehazing process. The inputs judged as clear will be directly output, while the hazy will be directed to the dehazing processing branch. Subsequently, the hazy inputs will be processed by multiple cascaded lightweight dehazing modules. The processed dehazing features will be fused adaptively with

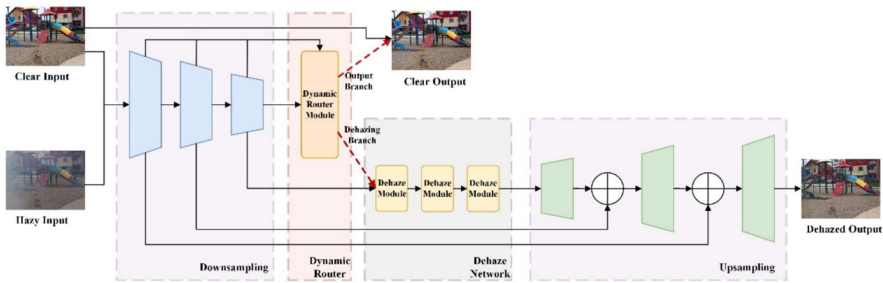


Fig. 1 Overall framework for the proposed network

the downsampled features before dehazing. Finally, the dehazing result will be reconstructed and generated through the upsampling module that combines the residual network and PixelShuffle.

3.2 Adaptive routing module

In most existing dehazing algorithms, the necessity of choosing an appropriate processing branch according to different input image types is often overlooked, which may lead to problems of incomplete dehazing or excessive dehazing in practical application scenarios. In fact, there is usually an obvious pixel-level difference between clear images and haze images in specific areas, which is particularly significant in real hazy images. To utilize this characteristic, we design an adaptive routing module based on multi-scale feature extraction. This module can analyze downsampled feature maps of different scales, score the input image, and determine the subsequent processing strategy accordingly. Specifically, if the score of the image is lower than the set threshold, we will directly output it without performing dehazing processing; if the score is higher than the threshold, the image will enter the dehazing processing branch for corresponding clarification processing, and finally generate the dehazed image. This adaptive strategy not only better meets the needs of different weather scenarios to improve dehazing efficiency, but also ensures higher-quality image reconstruction.

First, the downsampled features at each scale are subjected to global average pooling to effectively extract the key information of the features at each scale, and these pooled features are concatenated together to form a comprehensive multi-scale spliced feature F_{all} . This operation is shown in Eq. (1):

$$F_{\text{all}} = \text{Concat}([P(f_1), P(f_2), \dots, P(f_n)]) \quad (1)$$

where Concat is the concatenation operation, Pool is the global average pooling, and f_i is the i th downsampled feature map.

Then, the multi-scale spliced features F_{all} are fed into the two fully connected layers for further mapping process, which finally outputs a score that will be used for subsequent decision judgment. This operation is shown in Eq. (2):

$$\phi = \theta(\omega_2(\delta(\omega_1 \times F_{\text{all}} + b_1) + b_2)) \quad (2)$$

where θ and δ represent the sigmoid activation function and the ReLU activation function, ω_1 and ω_2 are the weight terms of the two fully connected layers, b_1 and b_2 are the bias terms of the two fully connected layers, and F_{all} is the multi-scale concatenated feature.

After training the adaptive routing module, a soft threshold score can be obtained. By setting a continuous threshold parameter between 0 and 1, the adaptive routing module can precisely adjust the processing ratio of hazy and clear images. As the threshold decreases, the network will perform dehazing processing on more images. Conversely, increasing the threshold means that more images will be directly output.

3.3 De-redundant feature attention Dehazing module

Inspired by the efficient feature attention (FA) dehazing module proposed by FFA-Net [19], AECR-Net [20] proves that simple modules can also achieve efficient dehazing effects by only using a few cascaded FA blocks. However, diverse redundant information in low-resolution features may mislead the attention to key features during dehazing processing. Therefore, we propose a lightweight de-redundant feature attention (DRFA) dehazing module. This module can more effectively focus on key features during dehazing and reduce the interference caused by redundant information. The structure of the DRFA dehazing module is shown in Fig. 2.

To reduce the negative impact of redundant information on local feature attention and the additional computational overhead caused by redundant information, we optimize based on the original FA dehazing module and replace regular convolution with SCConv convolution that reduces spatial and channel redundancy. SCConv consists of two sequentially placed units, the spatial reconstruction unit (SRU) and the channel reconstruction unit (CRU), which can reduce information redundancy in

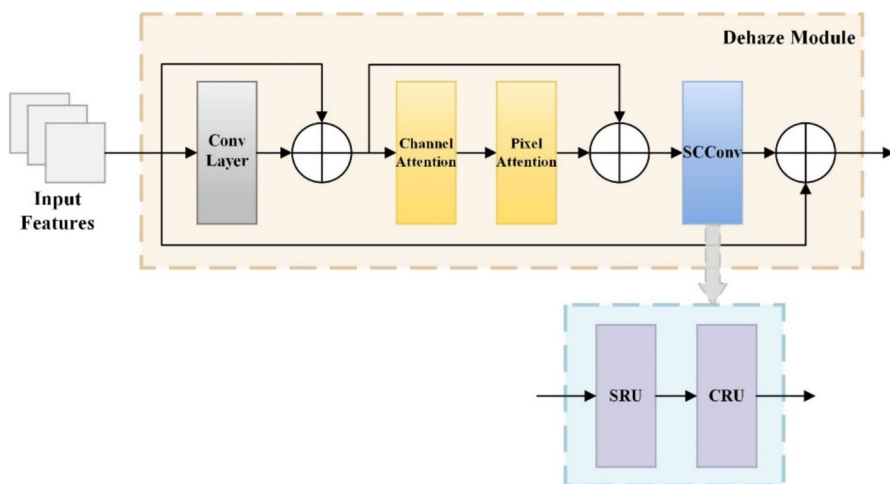


Fig. 2 Structure of the de-redundant feature attention (DRFA) dehazing module

both spatial and channel dimensions. It can both reduce the computational burden and facilitate the focus on key features, thus enhancing the model's ability to capture details.

While constructing the network, we reused the original input features several times and screened the complex features extracted by the attention mechanism through SCConv to remove the redundant information. Eventually, the unprocessed original input features are fused with the enhanced local features and output, which highlights the key local features in the feature map, and also retains the integrity of the original information, achieving significant suppression of haze and overall improvement of image quality.

3.4 Upsampling module

During image reconstruction, the choice of different upsampling algorithms can lead to different forms of representation of deep features after dimension space expansion. For instance, although commonly used interpolation algorithms can rapidly expand the spatial size, they rely too much on local pixel features and are not as effective in reconstructing image details. On the other hand, the more commonly used transposed convolution algorithm can achieve better reconstruction results through parameter learning; however, it is sensitive to hyperparameters such as convolution kernel size and step size, which is prone to the checkerboard effect in the generated image. To address the above problems, in the upsampling reconstruction stage, we propose an upsampling module that combines the convolution of residual structures with PixelShuffle to replace the commonly used transposed convolution for image restoration. This module can train the recovery of upsampled features at a lower cost of number of parameters and computational latency, and generates higher-quality dehazed reconstructed images.

The proposed upsampling module first increases the number of channels of the low-resolution feature map by 1×1 convolution for subsequent channel transformation operations. Then image resolution enhancement is performed using PixelShuffle, and a convolutional layer is added to learn the mapping and reconstruction information. Unlike the traditional onetime upsampling operation that directly restores the input image size, we use multiple enlargements of the resolution of the deep features after dehazing until they are restored to the original size. This method can highlight the key feature information to avoid losing too much image information during the upsampling process of oversize transformation, which leads to degradation of the quality of the dehazed reconstructed image. The specific operation is shown in Eqs. (3) and (4):

$$F_{\text{middle}} = \text{Pixel Shuffle}(\text{Conv}_1(F_{\text{in}})) \quad (3)$$

where F_{middle} is the intermediate feature after PixelShuffle upsampling, F_{in} is the input post-dehazing feature, Conv_1 is the 1×1 convolution used to increase the number of channels, and Pixel Shuffle is the upsampling algorithm; here, the size magnification is 2.

$$F_{\text{out}} = F_{\text{middle}} + \text{Conv}_2(F_{\text{middle}}) \quad (4)$$

where F_{middle} is the above intermediate feature, Conv_2 is a 3×3 convolution, which can increase the nonlinear expression capability of the network while ensuring that the number of output channels meets the output requirements, and F_{out} is the final output feature after upsampling.

3.5 Loss function

The loss function is mainly composed of three parts: image structure reconstruction loss, contrast loss and adaptive routing dynamic loss. Among them, the image structure reconstruction loss adopts the hybrid weighted loss of L1 loss and L2 loss, as expressed in Eq. (5):

$$\begin{aligned} L_{\text{Construction}} &= \lambda_1 * L_1 + \lambda_2 * L_2 \\ L_1 &= \|J - \text{Net}(I)\|_1 \\ L_2 &= \|J - \text{Net}(I)\|_2 \end{aligned} \quad (5)$$

where $L_{\text{Construction}}$ is the image structure reconstruction loss, L_1 and L_2 are the losses, I and J are the hazy image and the corresponding clear image, Net is the proposed dehazing network, and λ_1 and λ_2 are the weights corresponding to the L_1 and L_2 losses.

In the contrastive loss, to improve the detailed feature extraction ability of the model, ResNet101 is used to pre-train the model, as shown in Eq. (6):

$$L_{\text{Contrastive}} = \sum_{i=1}^{i=5} \omega_i * \frac{D(G_i(J), G_i(\text{Net}(I)))}{D(G_i(I), G_i(\text{Net}(I)))} \quad (6)$$

where D is the loss for evaluating the feature variability, here L_1 loss is chosen, G is the pre-trained model used for feature extraction, here ResNet101 is chosen, I and J are the hazy image and the corresponding clear image, ω is the weight coefficients on the different feature layers, and i is the indexes of the different feature layers.

In the adaptive routing dynamic loss design, we use the output variability between the direct output branch and the dehazing processing branch based on the type of input image to achieve the dynamic adjustment and learning of the module, as shown in Eq. (7):

$$\begin{aligned} L_{\text{dynamic}} &= \phi(\text{loss}_1 - \frac{\text{diff}}{2}) + (1 - \phi)(\text{loss}_2 + \frac{\text{diff}}{2}) \\ \text{diff} &= \text{Mean}(\text{loss}_1 - \text{loss}_2) \end{aligned} \quad (7)$$

where ϕ is the score of the output of the above adaptive routing module, loss_1 and loss_2 are the L_1 loss between the outputs of the direct output branch and the dehazing branch and the target image, respectively, and diff is a reconciliation loss term used to balance the inconsistency of the training difficulty of the two different tasks.

In summary, the total training loss L_{all} of our network is defined as shown in Eq. (8):

$$L_{\text{all}} = \alpha * L_{\text{Construction}} + \beta * L_{\text{Contrastive}} + \gamma * L_{\text{dynamic}} \quad (8)$$

where α , β , and γ are the weight terms of different losses, and $L_{\text{Construction}}$, $L_{\text{Contrastive}}$, and L_{dynamic} are the image structure reconstruction loss, contrast loss, and adaptive routing dynamic loss mentioned above, respectively.

4 Experiments

4.1 Datasets

We train and test on the RESIDE [21] dataset as well as NH-HAZE [22] and Dense-HAZE [23] datasets. Among them, the RESIDE dataset is a large-scale synthetic dataset, and its subset contains hazy images and corresponding clear images in different indoor and outdoor environments. Since we mainly focus on outdoor scenes, the outdoor training set (OTS) subset is selected as the training set, and 36,000 hazy–clear image pairs are extracted as training samples. In the testing phase, the effectiveness and generalization of the model are evaluated using the subset synthetic objective testing set-outdoor (SOTS-Outdoor). Concurrently, to enhance the model's capacity for dehazing in authentic scenarios, we have selected the real hazy datasets NH-HAZE and Dense-HAZE to augment the training dataset.

4.2 Experimental settings

In the training phase, the parameters of the experimental test system are shown in Table 1, and the parameter settings of the model are shown in Table 2. In setting the weight coefficients of the loss function, we adopt a dynamic adjustment strategy, which regards the training process of the model as a gradual evolution of multiple stages, in which the weight coefficients in the image structure reconstruction loss

Table 1 Parameters of the experimental test system

Equipment and environment	Type	Number
Server	SuperMicro 7048GR-TR	1
Operating system	Ubuntu 22.04 LTS	–
CPU	12th Gen Intel Core i7-12700 K	2
RAM	Samsung32G 2RX4 2400 T	4
GPU	NVIDIA GeForce RTX 3090	1
Development environment	Python 3.10	–
Development tools	Pycharm, VSCode	–
Framework and dependencies	Pytorch, OpenCV, etc	–

Table 2 Parameter settings of the training

Parameter	Description
Optimizer	Adam
Learning rate decay	$\beta_1 = 0.9, \beta_2 = 0.999$
Learning rate	$3e-4$
Learning rate decay	CosineAnnealingLR
Epoch	1000

are finally set to $\lambda_1 = 0.6$ and $\lambda_2 = 0.4$, respectively. In setting the weight coefficients of the loss function, we adopt a dynamic adjustment strategy, which regards the training process of the model as a gradual evolution of multiple stages, in which the weight coefficients in the image structure reconstruction loss are finally set to $a_1 = 0.6$ and $a_2 = 0.4$, respectively. At the early stage of training, we freeze the parameters of the adaptive routing module and train only on the dehazing processing branch. At this time, the weight of the image structure reconstruction loss is set to 0.8, while the weight of the contrast loss is set to 0.2. Finally, in the overall fine-tuning stage of the dehazing model, the weight of the image structure reconstruction loss is adjusted to 0.7, the weight of the contrast loss is adjusted to 0.15, and the weight of the adaptive routing loss is adjusted to 0.15.

It should be noted that in the subsequent comparison test between the dehazing effect and the model inference speed, since all the inputs are hazy pictures, we set the preset thresholds in the adaptive routing module of the model in the comparison experiment part to 0, i.e., if not otherwise specified, all the pictures will go through the dehazing processing branch.

4.3 Experimental result analysis

We use peak signal-to-noise ratio (PSNR) [24] and structural similarity (SSIM) [25] as objective evaluation metrics. In addition, we have conducted a subjective visual evaluation to further validate the effectiveness and reliability of the present algorithm in practical use. We have selected several representative dehazing algorithms for comparison, including DCP [8], DehazeNet [11], AOD-Net [12] algorithm, and several SOTA dehazing algorithms based on deep learning in recent years, including MSBDN [13], KDDN [14], AECD-Net [20], DeHamer [15], DEA-Net [26].

4.3.1 Results on synthetic hazy dataset

On the SOTS-Outdoor dataset, the dehazing performance of our method is compared with other SOTA algorithms, and the results are shown in Table 3. As seen from the values in Table 3, the PSNR and SSIM metrics of our method outperform most of the algorithms on the synthetic hazy dataset with 33.67 dB and 0.984, respectively. In addition, we also compare the image visual quality of different algorithms. Some of the results are shown in Fig. 3. It can be seen that the image recovered by DCP is dark, and there is a more obvious color distortion problem in

Table 3 Comparison results of PSNR and SSIM for each dehazing method on SOTS-Outdoor dataset

Method	PSNR	SSIM
DCP [8]	19.13	0.861
DehazeNet [11]	24.75	0.927
AOD-Net [12]	24.14	0.920
MSBDN [13]	34.81	0.986
DeHamer [15]	35.18	0.986
DEA-Net [26]	36.03	0.989
Ours	33.67	0.984

The best results for each method are displayed in bold

the sky area. DehazeNet and AOD-Net have the problem of incomplete dehazing, and there is still a certain color deviation in some areas. The dehazing effects of MSBDN, DeHamer, and our algorithms are clearer, and the degree of restoration of the dehazed image is very close to the clear image, showing a good visual effect.

4.3.2 Results on real hazy dataset

The test results on the real datasets NH-HAZE and Dense-HAZE are shown in Table 4. Our algorithm achieves the highest PSNR and the third highest SSIM score on NH-HAZE at 21.70 dB and 0.6952, respectively. On the Dense-HAZE dataset, we achieve the second highest PSNR and the front-runner SSIM score of 16.17 dB and 0.4538, which are only 0.45 dB and 0.1064 lower than the top-ranked DeHamer. To show the dehazing effect more intuitively, some of the dehazed results of each algorithm are shown in Figs. 4 and 5. It can be seen that the traditional DCP algorithms still face challenges on real datasets, and the images after dehazing show serious problems such as color distortion and artifacts. DehazeNet, AOD-Net, and MSBDN still have residual haze occlusion in the images after dehazing. In contrast, DeHamer and ours algorithm can effectively remove the haze and present clearer image. It should be specifically noted that DeHamer improves the dehazing performance by increasing the depth of the network, thus showing a better performance in quantitative indicators. In the follow-up, comparative experiments will be carried out for the inference efficiency of each algorithm to further verify the comprehensive effectiveness of different methods in practical deployment.

At the same time, we test and analyze the computational efficiency of the network on 256×256 size images, by quantifying and analyzing the number of model parameters, floating-point operations (FLOPs), and running time as evaluation indexes, and comparing and analyzing them with some algorithms based on deep learning in recent years, the results are shown in Table 5. The results show that our algorithm achieves the best performance in three evaluation metrics, has the highest inference efficiency, and can meet the requirements of real-time processing. It means that our algorithm can maintain a good dehazing effect while significantly reducing the number of parameters.



Fig. 3 Some of the dehazing results on SOTS-Outdoor datasets

In addition, to verify the adaptive processing ability of our algorithms in real scenarios, we especially compared different dehazing algorithms in the NH-HAZE dataset with mixed input of clear and hazy images. The results are shown in Table 6. It is specifically noted that due to the balanced number of hazy and clear samples in this experiment, the prior probabilities of the two classes are equal (i.e., $P(\text{hazy}) = P(\text{clear}) = 0.5$). Based on Bayesian decision theory, selecting 0.5 as the decision boundary minimizes the classification error rate. Additionally, the

Table 4 Comparison results of PSNR and SSIM for each dehazing method on real hazy dataset

Method	NH-HAZE		Dense-HAZE	
	PSNR	SSIM	PSNR	SSIM
DCP [8]	10.57	0.5196	10.06	0.3856
DehazeNet [11]	16.62	0.5238	13.84	0.4252
AOD-Net [12]	15.40	0.5693	13.14	0.4144
MSBDN [13]	19.23	0.7056	15.37	0.4858
KDDN [14]	17.39	0.5897	14.28	0.4074
AECR-Net [20]	19.88	0.7173	15.80	0.4660
DeHamer [15]	20.66	0.6844	16.62	0.5602
Ours	21.70	0.6952	16.17	0.4538

The best results for each method are displayed in bold

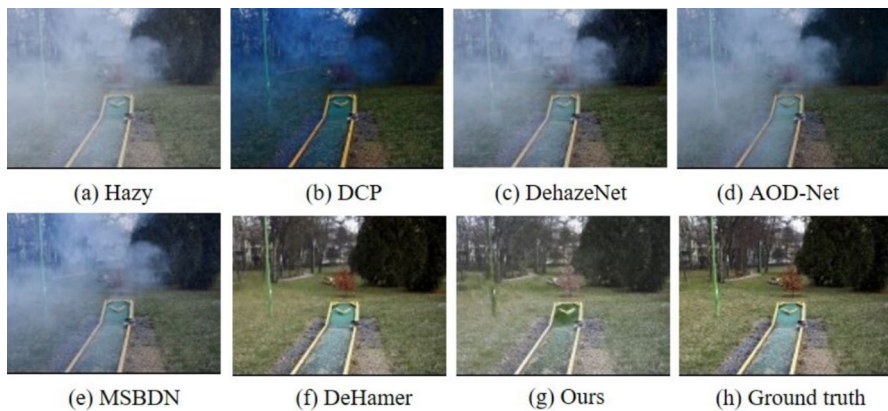
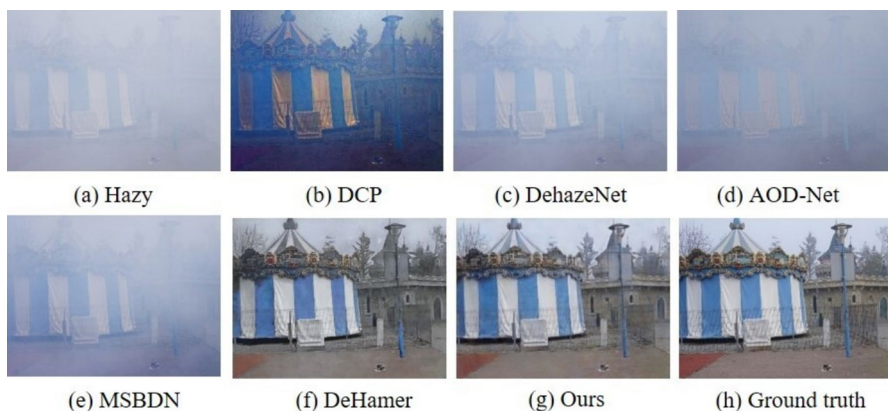
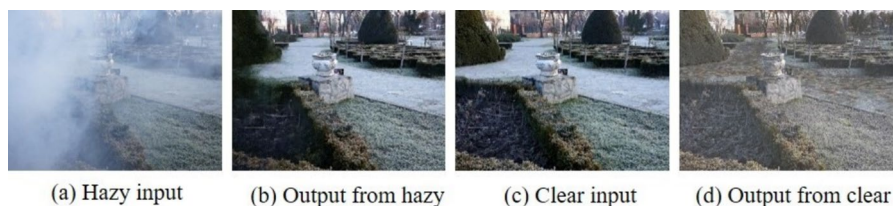
**Fig. 4** Some of the dehazing results on NH-HAZE dataset**Fig. 5** Some of the dehazing results on Dense-HAZE datasets

Table 5 Comparison results of inference efficiency metrics

Method	Params (M)	FLOPs(G)	Latency (ms)
MSBDN [13]	31.35	24.44	9.83
KDDN [14]	2.96	9.73	6.95
AECR-Net [20]	2.61	52.20	17.04
DeHamer [15]	132.4	48.93	14.12
DEA-Net [26]	3.65	32.23	7.09
Ours	0.89	8.93	4.81

Table 6 Dehazing results of each algorithm for different types of mixed input images

Method	Mixed dataset of clear- hazy images	
	PSNR	SSIM
DCP [8]	8.93	0.473
DehazeNet [11]	12.26	0.494
AOD-Net [12]	12.11	0.483
MSBDN [13]	15.14	0.511
KDDN [14]	15.22	0.507
AECR-Net [20]	16.39	0.535
DeHamer [15]	17.41	0.513
Ours	26.75	0.832

**Fig. 6** Some dehazing results of AECR-Net on NH-HAZE dataset for different types of pictures

sigmoid activation function maps the output of the adaptive routing module to the interval $[0, 1]$, with 0.5 serving as the symmetric midpoint of this interval to achieve equiprobable classification decisions, effectively partitioning the decision space between the two classes. Through the above theoretical analysis, to balance the uncertainty in the classification process and minimize entropy, the threshold of the adaptive routing selection module is set to 0.5 in this experiment.

As shown in Fig. 6, since the other comparison algorithms lack adaptive routing mechanisms, they process the input images indiscriminately, which may lead to over-processing of the input clear images, thus degrading the quality of the images. In the case of mixed-type image input, our algorithm significantly reduces the image degradation problem that may be caused by the dehazing of clear images, thus

achieving better dehazing results on mixed-type images. Compared with others, our algorithm gets the highest scores in both PSNR and SSIM indicators, which fully proves its adaptive processing ability in real application scenarios, and can better meet the needs of different scenarios.

4.4 Ablation experiment

In order to deeply explore the influence of the upsampling algorithm on the effect of dehazing reconstructed images, we randomly selected 10 training images in the Dense-HAZE and NH-HAZE datasets and constructed a hybrid dataset for ablation experiments. In the experiments, we will use the model of conventional transposed convolution as the Baseline. Then, we replace the upsampling algorithm with the interpolation algorithm and the proposed upsampling algorithm with the residual structure in turn, respectively. The experimental results are shown in Table 7. The results of ablation experiments show that the parameter learning-based upsampling algorithms (i.e., transposed convolution and the proposed upsampling algorithm) outperform the interpolated upsampling algorithms in the evaluation metrics of the dehazing results. In addition, the proposed upsampling algorithm, through the combination of residual structure and PixelShuffle, achieves a better dehazing reconstruction by improving the PSNR and SSIM by 0.47 dB and 0.06, respectively, compared to the method using transposed convolution.

5 Conclusion

Aiming at the problems of poor input adaptation and high computational cost faced in the process of dehazing in outdoor scenes, we propose a real-time image dehazing algorithm based on adaptive routing. Specifically, we first design an adaptive routing module based on multi-scale feature extraction, which analyzes the downsampled feature maps at different scales and scores the features of the input images by using the adaptive routing mechanism to determine whether they need to enter the dehazing process. For images judged as clear, the algorithm will output them directly, while hazy images will be guided to the dehazing processing branch. In addition, we propose a DRFA module. This module is optimized based on the traditional FA module and introduces the SCConv convolution to replace the regular convolution, aiming at reducing spatial and channel redundancy. Meanwhile, by incorporating residual connections, key feature information

Table 7 Comparative results using different upsampling methods on mixed datasets

Method	PSNR	SSIM
Baseline	16.89	0.53
Baseline + interpolation algorithm	16.38	0.51
Baseline + the proposed upsampling algorithm	17.36	0.59

is retained while reducing the impact of redundant information. Finally, we propose an upsampling module that combines residual networks and PixelShuffle to achieve effective recovery of upsampled features with low parametric and computational costs, while ensuring high-quality outputs of dehazed reconstructed images. The experimental results show that the proposed dehazing algorithm exhibits better dehazing performance on both synthetic hazy datasets and real hazy datasets, and improves the real-time performance of the algorithm without significantly reducing the dehazing effect. It also has a certain adaptive processing capability, which can better meet the needs of different scenarios. On SOTS-Outdoor, we achieved high PSNR score (33.67dB) and SSIM score (0.984). On NH-HAZE, the highest PSNR score (21.70dB) and the third highest SSIM score (0.6952) were obtained, and the second highest PSNR (16.17dB) and the top SSIM score (0.4538) were achieved on Dense-HAZE, which were only 0.45dB lower than the top-ranked DeHamer, respectively, and 0.1064. In the case of clear image and haze image mixed input, we obtained the highest PSNR (26.75dB) and SSIM score (0.832), much higher than the scores of other comparison algorithms. The proposed algorithm shows significant advantages in terms of operation efficiency. During the inference efficiency test of 256×256 -pixel images, our algorithm has a computational cost of 8.93G FLOPs, and the inference time consumed is only 4.81ms, which makes it very suitable for edge devices and tasks with high real-time requirements. Although our algorithm significantly reduces computational overhead, it inevitably compromises dehazing performance to some extent. Therefore, in future work, we plan to further improve the image restoration quality by designing more efficient dehazing modules.

Acknowledgements This work was supported by the Major Technology Innovation of Hubei Province (2022BEC026).

Author contributions K.M.: Conceptualization, Data curation, Validation, Software, Writing—review & editing. S.Y.: Methodology, Data curation, Formal analysis, Validation, Writing—original draft. Z.L.: Data curation, Investigation, Software, Validation, Writing—review & editing. S.L.: Data curation, Visualization, Software, Validation, Writing—review & editing. M.W.: Data curation, Supervision, Writing—review & editing. H.L.: Data curation, Supervision, Writing—review & editing. L.Y.: Data curation, Funding acquisition, Resources, Supervision, Writing—review & editing.

Funding Financial support was provided by the Major Technology Innovation of Hubei Province.

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest There is no conflict of interest to disclose.

Ethical approval Not applicable.

Consent to participate Not applicable.

Consent to publication Not applicable.

References

- Ren W, Pan J, Zhang H et al (2020) Single image dehazing via multi-scale convolutional neural networks with holistic edges. *Int J Comput Vision* 128(1):240–259. <https://doi.org/10.1007/s11263-019-01235-8>
- Pan J, Dong J, Liu Y et al (2021) Physics-based generative adversarial models for image restoration and beyond. *IEEE Trans Pattern Anal Mach Intell* 43(7):2449–2462. <https://doi.org/10.1109/TPAMI.2020.2969348>
- Bai H, Pan J, Xiang X et al (2022) Self-guided image dehazing using progressive feature fusion. *IEEE Trans Image Process* 31:1217–1229. <https://doi.org/10.1109/TIP.2022.3140609>
- Cantor A (1978) Optics of the atmosphere—scattering by molecules and particles. *IEEE J Quantum Electron* 14(9):698–699. <https://doi.org/10.1109/JQE.1978.1069864>
- Xu Z, Liu X, Chen X (2009) Fog removal from video sequences using contrast limited adaptive histogram equalization. In: 2009 International Conference on Computational Intelligence and Software Engineering, pp 1–4. <https://doi.org/10.1109/CISE.2009.5366207>
- Jobson DJ, Rahman Z, Woodell GA (1997) A multiscale retinex for bridging the gap between color images and the human observation of scenes. *IEEE Trans Image Process* 6(7):965–976. <https://doi.org/10.1109/83.597272>
- Yu L, Liu X, Liu G et al. (2015) A new dehazing algorithm based on overlapped sub-block homomorphic filtering. In: 18th International Conference on Machine Vision (ICMV 2015), pp 987502. <https://doi.org/10.1117/12.2228467>
- He K, Sun J, Tang X (2009) Single image haze removal using dark channel prior. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp 1956–1963. <https://doi.org/10.1109/CVPR.2009.5206515>
- Zhu Q, Mai J, Shao L (2015) A fast single image haze removal algorithm using color attenuation prior. *IEEE Trans Image Process* 24(11):3522–3533. <https://doi.org/10.1109/TIP.2015.2446191>
- Berman D, Treibitz T, Avidan S (2016) Non-local image dehazing. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp 1674–1682. <https://doi.org/10.1109/CVPR.2016.185>
- Cai B, Xu X, Jia K et al (2016) DehazeNet: An end-to-end system for single image haze removal. *IEEE Trans Image Process* 25(11):5187–5198. <https://doi.org/10.1109/TIP.2016.2598681>
- Li B, Peng X, Wang Z et al. (2017) AOD-Net: All-in-one dehazing network. In: Proceedings of the IEEE International Conference on Computer Vision (ICCV), pp 4780–4788. <https://doi.org/10.1109/ICCV.2017.511>
- Dong H, Pan J, Xiang L et al. (2020) Multi-scale boosted dehazing network with dense feature fusion. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp 2154–2164. <https://doi.org/10.1109/CVPR42600.2020.00223>
- Hong M, Xie Y, Li C et al. (2020) Distilling image dehazing with heterogeneous task imitation. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp 3459–3468. <https://doi.org/10.1109/CVPR42600.2020.00352>
- Guo C, Yan Q, Anwar S et al. (2022) Image dehazing transformer with transmission-aware 3D position embedding. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp 5802–5810. <https://doi.org/10.1109/CVPR52688.2022.00572>
- Wang T, Zhao L, Huang P et al (2021) Haze concentration adaptive network for image dehazing. *Neurocomputing* 439:75–85. <https://doi.org/10.1016/j.neucom.2021.01.042>
- Yu Z, Peng J (2024) Adaptive multi-information distillation network for image dehazing. *Multimed Tools Appl* 83:18407–18426. <https://doi.org/10.1007/s11042-023-15091-9>
- Li X, Hua Z, Li J (2023) Attention-based adaptive feature selection for multi-stage image dehazing. *Vis Comput* 39:663–678. <https://doi.org/10.1007/s00371-021-02365-2>
- Qin X, Wang Z, Bai Y et al (2020) FFA-net: feature fusion attention network for single image dehazing. *Proc AAAI Conf Artif Intell* 34(07):11908–11915. <https://doi.org/10.1609/aaai.v34i07.6865>
- Wu H, Qu Y, Lin S et al. (2021) Contrastive learning for compact single image dehazing. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp 10546–10555. <https://doi.org/10.1109/CVPR46437.2021.01041>
- Li B, Ren W, Fu D et al (2019) Benchmarking single-image dehazing and beyond. *IEEE Trans Image Process* 28(1):492–505. <https://doi.org/10.1109/TIP.2018.2867951>

22. Ancuti CO, Ancuti C, Timofte R (2020) NH-HAZE: An image dehazing benchmark with non-homogeneous hazy and haze-free images. In: 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), pp 1798–1805. <https://doi.org/10.1109/CVPRW50498.2020.00230>
23. Ancuti CO, Ancuti C, Sbert M et al. (2019) Dense-Haze: A benchmark for image dehazing with dense-haze and haze-free images. In: 2019 IEEE International Conference on Image Processing (ICIP), pp 1014–1018. <https://doi.org/10.1109/ICIP2019.8803046>
24. Sheikh HR, Sabir MF, Bovik AC (2006) A statistical evaluation of recent full reference image quality assessment algorithms. *IEEE Trans Image Process* 15(11):3440–3451. <https://doi.org/10.1109/TIP.2006.881959>
25. Wang Z, Bovik AC, Sheikh HR et al (2004) Image quality assessment: from error visibility to structural similarity. *IEEE Trans Image Process* 13(4):600–612. <https://doi.org/10.1109/TIP.2003.819861>
26. Chen Z, He Z, Lu Z-M (2024) DEA-Net: Single image dehazing based on detail-enhanced convolution and content-guided attention. *IEEE Trans Image Process* 33:1002–1015. <https://doi.org/10.1109/TIP.2024.3354108>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.